

... — Glass Law — ...

∴ Boyle's Law:-

$$(V \propto \frac{1}{P}) \text{ (Constant } T, n)$$

∴ Charles Law:-

$$(V \propto T) \text{ (Constant } P, n)$$

∴ Avogadro's Law:-

$$(V \propto n) \text{ (Constant } T, P)$$

Ideal gas:-

"The gas that obey all gas laws and have no forces of attraction among gas molecule called Ideal gas."

❖ Ideal gas not exist in nature.

At high T and low P gasses are ideal.

❖ Because at high Temperature gas molecules expand and weaker forces of attraction present b/w gas molecules.

❖ Similarly at low P the gas molecule are not fully compressed and force of attraction are not present so gases are ideal at low P .

Ideal general gas equation:-

$$PV = nRT$$

R = general ideal gas constant.

its value is $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.

Non-ideal gases:-

Non-ideal gasses are also called real gases. All gases in nature are non ideal gases and not obey gas law.

- Force of attraction present b/w gas molecules.

- Low Temperature

- Pressure increase than gases are non-ideal.

- Because at low temp and high P gas molecules are compressed and Force of attraction present b/w gas molecule to gas are non-ideal.

Non-ideal gas equation:-

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

P = Pressure

V = Volume

a, b = Vander walls constants.

b = excluded volume / Co volume / effective

Vol, compressible volume.

→ The value of a and b (depend)^s depressed upon:-

- Temperature
- Nature of gas
- Molecular mass

These factors build up force of attraction.

Vander waal's equation:-

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

units of vander waal's ^{constant} equation:-

$$\left(P + \frac{an^2}{V^2}\right)$$

unit of a :-

$$a = \frac{PV^2}{n^2}$$

$$a = \frac{Nm^2(m^3)^2}{mol^2} \Rightarrow \frac{Nm^2 m^6}{mol^2}$$

$$a = Nm^4 mol^2$$

units of b :-

$$b = V - nb$$

$$b = \frac{V}{n}$$

$$b = \frac{m^3}{mol}$$

$$b = m^3 mol^{-1}$$

Virial Equation

History/Discovery:-

⇒ Virial equation proposed by Heikh Kamerlingh onnes.

⇒ He was Dutch Physicist and noble Laurate.

⇒ Equation is called Kamerlingh onnes virial equation.

Noble Prize:-

He awarded noble Prize in 1930.

Reason of noble Prize:-

- He awarded noble Prize not of virial equation but he awarded noble Prize because of:-

⇒ Liquification of He gas in 1908.

⇒ Discover Superconductivity in 1911.

Liquification:-

“The process of conversion of gas into Liquid Liquification.”

⇒ He and H_2 gases were not

Liquifying because of:-

⇒ There small size.

⇒ no force of attraction present b/w

gas molecules.

Super-Conductivity:-

Some material that are Super conductors that have maximum conductivity & zero resistance.

Virial means $\Rightarrow$ Force of attraction blw gass molecules.

Beacuse non-ideal gas molecules have Force of attraction.

Derivation:-

Ideal gas equation is:-

$$PV = nRT$$

For one mole ($n=1$)

$$PV = RT$$

$$Z = \frac{PV}{RT}$$

Z = Compressibility Factors that is defined as:-

"it is deviation of real gas from ideal behaviour."

Z = Molar volume of real gas

Molar volume of an ideal gas

For ideal gas ($Z=1$)

According to vander waal's eq:-

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

where a and b are vander waals constant.

- a = Coefficient of attraction.
- b = excluded vol effective vol, co-volume, compressible volume.

Units of a and b :-

units of a :- $P + \frac{an^2}{V^2}$

$$a = \frac{PV^2}{n^2} \Rightarrow \frac{Nm^{-2}(m^3)^2}{mol^2}$$

$$a = \frac{Nm^{-2}m^6}{mol^2} \Rightarrow \boxed{a = Nm^4mol^{-2}}$$

units of b :-

$$b = V - nb$$

$$b = \frac{V}{n}$$

$$b = \frac{m^3}{mol}$$

$$\boxed{b = m^3mol^{-1}}$$

For real gases:-

$$Z = 1 + B'P + C'P^2 + D'P^3 \quad \text{--- (1)}$$

For ideal gas.

$Z = 1^{st}$ virial Co-efficient.

$B' = 2^{nd}$ " " "

$C' = 3^{rd}$ " " "

$D' = 4^{th}$ " " " "

At low P (less than 10 atm)

C' and D' because negligible

So eq 1 becomes:

$$Z = 1 + B'P \quad \text{--- (2)}$$

$\frac{PV}{RT} = 1$ So, value of P can be written as:-

$$P = \frac{RT}{V}$$

by putting the value of P in eq (2)

$$Z = 1 + B'\left(\frac{RT}{V}\right) + C'\left(\frac{RT}{V}\right)^2 + D'\left(\frac{RT}{V}\right)^3 \quad \text{--- (3)}$$

if:-

$$B'RT = B$$

$$C'(RT)^2 = C$$

$$D'(RT)^3 = D$$

The eq (3) becomes:-

According to vander waal's eq:-

$$\left(\frac{P + \frac{an^2}{V^2}}{V} \right) (V - nb) = nRT$$

where a and b are vander waals constant.

- a = Coefficient of attraction.
- b = excluded vol, effective vol, co-volume, compressible volume.

Units of a and b :-

units of a :- $\frac{P + \frac{an^2}{V^2}}{V}$

$$a = \frac{PV^2}{n^2} \Rightarrow \frac{Nm^{-2}(m^3)^2}{mol^2}$$

$$a = \frac{Nm^{-2}m^6}{mol^2} \Rightarrow \boxed{a = Nm^4mol^{-2}}$$

Units of b :-

$$b = V - nb$$

$$b = \frac{V}{n}$$

$$b = \frac{m^3}{mol}$$

$$\boxed{b = m^3mol^{-1}}$$

For real gases:-

$$Z = 1 + B'P + C'P^2 + D'P^3 \quad \text{--- (1)}$$

For ideal gas.

Z = 1st virial Co-efficient.

B' = 2nd " " "

C' = 3rd " " "

D' = 4th " " " "

At low P (less than 10 atm)

C' and D' because negligible

So eq 1 becomes:

$$Z = 1 + B'P \quad \text{--- (2)}$$

$\frac{PV}{RT} = 1$ So, value of P can be written as:-

$$P = \frac{RT}{V}$$

by putting the value of P in eq (2)

$$Z = 1 + B'\left(\frac{RT}{V}\right) + C'\left(\frac{RT}{V}\right)^2 + D'\left(\frac{RT}{V}\right)^3 \quad \text{--- (3)}$$

if:-

$$B'RT = B$$

$$C'(RT)^2 = C$$

$$D'(RT)^3 = D$$

The eq (3) becomes:-

$$Z = 1 + \frac{B}{V} + \frac{C}{V^2} + \frac{D}{V^3} \quad \text{--- 4}$$

This is real equation.

$B \gg C \gg D$.

eq (4) becomes.

$$Z = 1 + \frac{B}{V} \quad \text{--- 5}$$

2nd virial Co-efficient depends on Temperature.

• Low temp $\rightarrow$ B value (-)

• High Temp $\rightarrow$ B value (+)

Boyle's Temp:-

• At some time by changing temp

$B=0$

• The temp at which $B=0$ this temp called Boyle's temp and this obey Boyle's law.

• This gas obey Boyle's law is ideal gas.

Derivation of virial Equation from vander's waal's eq:-

According to vander's waal's eq.

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

Take:-

$$-\frac{V}{V-b}$$

For 1 mole:- $n=1$

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

"÷" by numerator or denominator by "V"

$$\frac{P + \frac{a}{V^2}}{V - b} = \frac{RT}{V}$$

$$= \frac{V/V}{(V-b)/V}$$

$$\frac{P}{V - b} = \frac{RT}{V^2} - \frac{a}{V^3} \quad \text{--- i}$$

$$= \frac{1}{\frac{V-b}{V}} \Rightarrow \frac{1}{1 - \frac{b}{V}}$$

Multiply with $\left(\frac{V}{RT}\right)$ on B.S From Taylor's Series:-

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$\frac{P \cdot V}{RT} = \left(\frac{RT}{V-b} - \frac{a}{V^2}\right) \frac{V}{RT}$$

$$\frac{1}{1 - \frac{b}{V}} = 1 + \frac{b}{V} + \frac{b^2}{V^2} + \frac{b^3}{V^3} + \dots$$

$$\frac{PV}{RT} = \frac{RT}{V-b} \cdot \frac{V}{RT} - \frac{a}{V^2} \cdot \frac{V}{RT}$$

because $\frac{b}{V} < 1$ so,

$$\frac{PV}{RT} = \frac{V}{V-b} - \frac{a}{VRT}$$

higher terms can be ignore.

• $\frac{PV}{RT} = Z \Rightarrow$ Compressibility factor.

$$\frac{1}{1 - \frac{b}{V}} = 1 + \frac{b}{V}$$

$$Z = \frac{V}{V-b} - \frac{a}{VRT} \quad \text{--- ii}$$

Put the value in eq ii

$$Z = 1 + \frac{b}{V} - \frac{a}{VRT} \quad \text{--- iii}$$

From eq. no 5

$$Z = 1 + \frac{B}{V} \Rightarrow Z = 1 + \frac{1}{V} B$$

From eq. no iii

$$Z = 1 + \frac{b}{V} - \frac{a}{VRT}$$

$$Z = 1 + \frac{1}{V} \left[b - \frac{a}{RT} \right] \quad \text{--- iv}$$

Comparing eq. no. iv & 5

$$B = b - \frac{a}{RT}$$

$\Rightarrow B$ include Both attractive and repulsive forces.

$\Rightarrow$ virial coefficient depends on nature of gas Temp & pressure.

At low pressure:

Only 1st virial coefficient is valid.

At high P:

2nd virial coefficient is also valid.

At constant Temp:-

virial coefficient are constant.

$\Rightarrow$ 1st virial coefficient is positive.

$\Rightarrow$ 2nd virial coefficient is negative (low p)

• At Boyle's Temp:-

Attraction forces equals to repulsive force. $B=0$.

$\Rightarrow B$ is due to molecular pair attraction

$\Rightarrow$ Higher virial coefficient are due to higher interaction.

At:

$$B = b - \frac{a}{RT}$$

At Boyle's Temp:-

$$B = 0$$

$$0 = b - \frac{a}{RT}$$

$$b = \frac{a}{RT}$$

$$T = \frac{a}{Rb} \rightarrow \text{Boyle's Temperature equation.}$$

$\Rightarrow$ At Boyle's Temp gas behave ideal.

$\Rightarrow$ At $b > \frac{a}{RT}$ (+ve deviation)

$\Rightarrow$ Effect of molecular size denominator.

$\Rightarrow$ At $b < \frac{a}{RT}$ (-ve deviation)

Determination of virial Coefficient:-

Ideal Gas equation for n^{th} of moles:-

$$PV = nRT$$

$$Z = \frac{PV}{nRT} = 1$$

Z = Compressibility factor.

$\Rightarrow$ For ideal gas $Z=1$:-

$\Rightarrow$ For non-ideal gas:-

$$Z = 1 + B'P + C'P^2 + D'P^3 \quad \text{--- ii}$$

$$PV = nRT$$

$$P = \frac{nRT}{V}$$

Put the value of " P " in eq ii

$$Z = 1 + B'\left(\frac{nRT}{V}\right) + C'\left(\frac{nRT}{V}\right)^2 + D'\left(\frac{nRT}{V}\right)^3$$

$$Z = 1 + \frac{B'nRT}{V} + \frac{C'n^2R^2T^2}{V^2} + \frac{D'n^3R^3T^3}{V^3} \quad \text{--- (iii)}$$

$$B'RT = B, \quad C'(nRT)^2 = C, \quad D'(RT)^3 = D$$

Put the values in eq iii

$$Z = 1 + \frac{Bn}{V} + \frac{Cn^2}{V^2} + \frac{Dn^3}{V^3} \quad \text{--- iv}$$

$$\text{Density} = \frac{M}{V}$$

$$\text{Molar density} = \frac{\text{moles}}{V}$$

$$d = \rho$$

Put the value " $d = \rho$ " in eq iv

$$Z = 1 + Bd + Cd^2 + Dd^3$$

$$\therefore Z = \frac{PV}{nRT}$$

Put the value of Z :-

$$\frac{PV}{nRT} = 1 + Bd + Cd^2 + Dd^3$$

$$\frac{P}{dRT} = 1 + Bd + Cd^2 + Dd^3 \quad \therefore d = \rho \Rightarrow \frac{1}{d} = \frac{1}{\rho}$$

$$\frac{P}{dRT} - 1 = Bd + Cd^2 + Dd^3$$

By ignoring the coefficient of " d^3 "

$$\frac{P}{dRT} - 1 = Bd + Cd^2 \quad \text{--- v}$$

divided by molar density on B.S of

eq v

$$\frac{1}{d} \left(\frac{P}{dRT} - 1 \right) = \frac{1}{d} (Bd + Cd^2)$$

$$\frac{1}{d} \left(\frac{P}{dRT} - 1 \right) = \frac{1}{d} \cdot Bd + \frac{1}{d} \cdot Cd^2$$

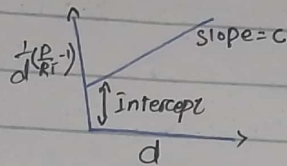
$$\frac{1}{d} \left(\frac{P}{dRT} - 1 \right) = B + Cd$$

This is final eq.

⇒ This is equation of straight line.

$$\underbrace{\frac{1}{d} \left(\frac{P}{RT} - 1 \right)}_y = \underbrace{B}_{\text{Intercept}} + \underbrace{c}_{\text{slope}/x} d$$

$$\text{Slope} = \frac{y}{x}$$



Maxwell's Law of Molecular Velocities:-

Molecular velocities:-

- Gas molecules move randomly or freely and collide with each other.
 - Due to Random motion their velocities and direction changes many times per second.
 - Liquid and Solid molecules don't move freely b/w their molecules. due to attractive forces.
 - Individual molecules velocities and K.E changes.
- At constant temp:-
Their velocities is also constant

and their K.E is also constant.

Elastic Collision:-

- These collisions are perfectly elastic i.e. there is no increase/decrease in energy of system.
- it means total energy of the system remains same after collision.
- Absolute value/Individual value of velocities and energy of gas molecule can not be determined because one mole of gas molecule contains avogadro's no.
- Velocities of gas molecule can be expressed as:-
 - Mean velocity
 - Root mean square velocity
 - Most Probable velocity

(i)

Mean velocity:-

"Sum of velocities of all gas molecules dividing by total no. of gas molecules"

Suppose:-

No. of molecules having velocity $C_1 = N_1$
 " " " " $C_2 = N_2$

No. of molecules having velocity $C_3 = N_3$

$$\bar{C} = \frac{C_1 N_1 + C_2 N_2 + C_3 N_3 + \dots}{N}$$

From Absolute value of velocities and energy of gas molecules can not be determined individually.

So, the Maxwell and Boltzmann gave the theory together.

From Maxwell-Boltzmann law of

Distribution of velocities-

Mean velocity can be determined as:-

$$\bar{C} = \sqrt{\frac{8R\bar{T}}{\pi M}}$$

R = General gas constant.

$\bar{T}$ = Temperature

M = Molecular mass of gas.

MCQ's Question:-

$$\bar{C} \propto \sqrt{\bar{T}}$$

$$\bar{C} \propto \frac{1}{\sqrt{M}}$$

(ii)

Root mean square velocity:-

$$\bar{C}^2 = \frac{N_1 C_1^2 + N_2 C_2^2 + N_3 C_3^2 + \dots}{N}$$

$$\sqrt{\bar{C}^2} = C_{rms} = \sqrt{\frac{N_1 C_1^2 + N_2 C_2^2 + N_3 C_3^2 + \dots}{N}}$$

Maxwell-Boltzmann Law:-

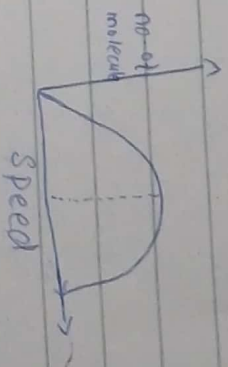
$$C_{rms} = \sqrt{\frac{3R\bar{T}}{M}}$$

(iii)

Most Probable velocity:-

"velocity of majority of gas molecules is called most Probable velocity."

$$C_{mp} = \sqrt{\frac{2R\bar{T}}{M}}$$



Distribution of molecular velocities of gasses:-

"This is a phenomenon in which molecules of same gas move with different velocities under same condition."

Types:-

- Narrow Distribution of velocity
- Broad Distribution of velocity

(i)

Narrow Distribution:-

If there is minor difference of velocities under same condition is called narrow distribution of velocity.

Broad Distribution:- (ii)

If there is major difference of velocities under same conditions is called broad distribution of velocity.

Explanation:-

- ⇒ Suppose we have one mole of gas molecules enclosed in a container.
- ⇒ Gas molecules move in different direction with different velocities.
- ⇒ Gas molecules undergo elastic collision.

⇒ K.E and velocity of molecule

gas molecule increase/decrease but there is no net loss of energy.

⇒ The average K.E, average velocity or root mean square velocity depends on Temperature.

• At constant temp:-

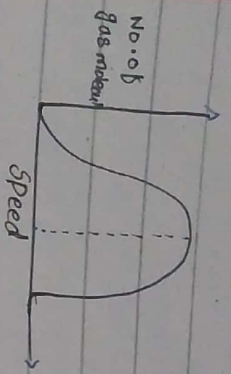
⇒ The average K.E, Average velocity or root mean square velocity are constant.

⇒ Gas molecules move in different directions with different velocities.

⇒ velocities of all molecules are not same because velocity changes many times per second due to collision.

⇒ velocities of gas molecules are in range of $-\infty$ to $+\infty$.

• According to James Clerk Maxwell and Ludwig Edward Boltzmann in 1860s at particular temp, fraction of molecules having particular velocities range remains constant.

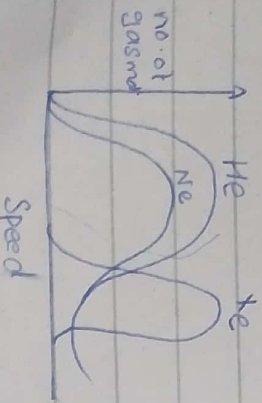
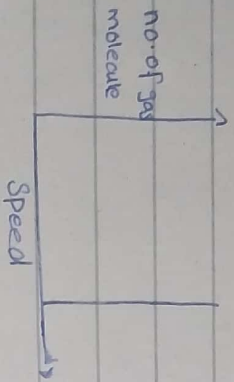


- The molecules which have low no. of gas molecules their speed is also low.

- The molecule which have intermediate no. of gas molecules their speed is also intermediate.

Ideal Case:-

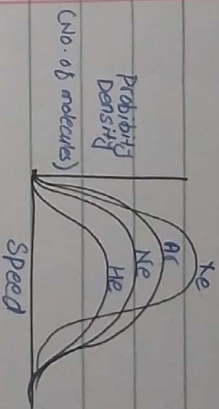
- All gas molecules have same velocity.



- if their molar mass of molecules is

increase than they have narrow distribution of velocity.

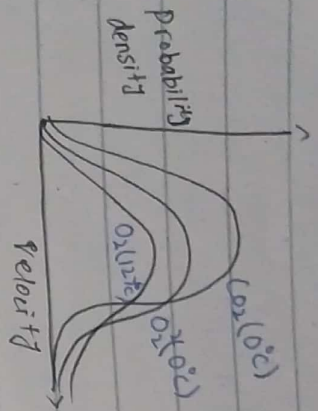
- if their molar mass of molecules is decrease than they have broad distribution of velocity.



⇒ For gases of lower masses peaks are shift towards right and downward.

⇒ Curve becomes flat. For gases of lower molar masses showing that lighter gas molecules move with greater speed than heavier one.

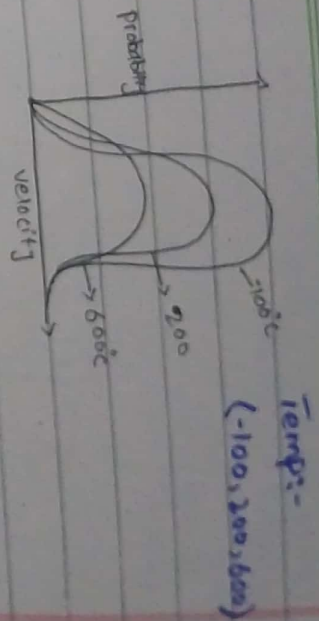
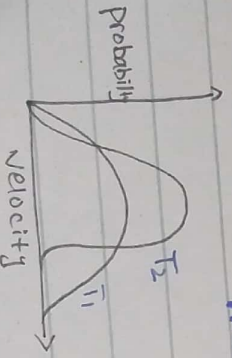
⇒ So, it is evident that lighter molecules have broad distribution velocity.
 ⇒ heavier molecules have narrow distribution.



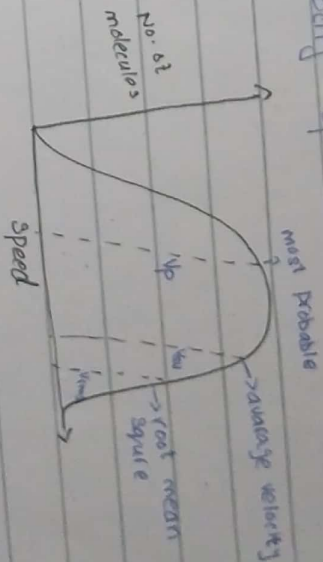
- The Peaks of Curves show most Probable velocity.
- At high temperature Peaks shift downward and towards right.
- Curves becomes flat showing that at high temp. greater no. of molecules are moving with high speed. So, there is more exchange of speed b/w gas molecule.
- Low temp $\rightarrow$ Narrow distribution
- High temp $\rightarrow$ Broad distribution

At:-

$$T_2 > T_1$$



- if the temperature is increase than the motion of molecules are also increase and than the velocity of Distribution becomes broad.



Derivation of expression for most Probable velocity (C_{mp}) from Maxwell Law of velocity distributions:

$$C_{mp} = \sqrt{\frac{2RT}{M}}$$

According to Maxwell-Boltzmann distribution Law:-

i) no. of gas molecules (dN) having specific velocity in the range of " C " and " $C+dc$ " is constant at constant temperature.

ii) Fraction of gas molecules ($\frac{dN}{N}$ or $f(c)dc$) having specific velocity in the range of " C " and " $C+dc$ " is constant at constant temperature.

iii) Probability of gas molecules ($\frac{1}{N} \frac{dN}{dc}$ or $f(c)$) having specific velocity in the range of " C " and " $C+dc$ " is constant at constant temp.

$$f(c) = 4\pi c^2 \left[\frac{M}{2\pi RT} \right]^{3/2} \cdot e^{-mc^2/2RT} \cdot \frac{1}{c}$$

$$f(c) = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2} \cdot c^2 e^{-mc^2/2RT}$$

$$\bullet \text{ Constant} = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2}$$

$$\bullet \text{ variable} = c^2 e^{-mc^2/2RT}$$

So eq (1) is:

$$\Rightarrow c^2 = \text{pre-exponential factor}$$

$$\Rightarrow e^{-mc^2/2RT} = \text{exponential factor}$$

- Pre-exponential factor (c^2) increase with Probability ($f(c)$).
- Exponential factor ($e^{-mc^2/2RT}$) decrease with Probability ($f(c)$).
- At higher value of C :
Pre-exponential factor is dominant and $f(c)$ increase.
- At lower value of C :
exponential factor is dominant and $f(c)$ decrease.
- At intermediate value of C :
- Both exponential & pre-exponential factors balance each other and Probability becomes constant.
- Fraction of gas molecules having velocity in the range of " C " and is constant at Constant temperature.

Derivation of constant is zero.

$$\frac{d}{dc} f(c) = 0$$

Putting $f(c)$ value from eq no 1:-

$$\frac{d}{dc} \left[4\pi c^2 \left(\frac{M}{2\pi R T} \right)^{3/2} \cdot e^{-mc^2/2RT} \right] = 0$$

$$4\pi \left[\frac{M}{2\pi R T} \right]^{3/2} \cdot \frac{d}{dc} c^2 e^{-mc^2/2RT} = 0 \quad \text{--- 2}$$

$$\therefore \frac{d}{dc} AB = A \frac{d}{dc} B + B \frac{d}{dc} A$$

$$\frac{d}{dc} c^2 e^{-mc^2/2RT} = c^2 \frac{d}{dc} e^{-mc^2/2RT} + e^{-mc^2/2RT} \frac{d}{dc} c^2$$

$$\therefore \frac{d}{dc} c^2 = 2c^{2-1} = 2c$$

$$\therefore \frac{d}{dc} \left(\frac{-Mc^2}{2RT} \right) = -\frac{M}{2RT} \frac{d}{dc} c^2 = -\frac{M}{2RT} \cdot 2c = -\frac{2Mc}{2RT}$$

$$\frac{d}{dc} c^2 e^{-mc^2/2RT} = c^2 e^{-mc^2/2RT} \left(\frac{-2Mc}{2RT} \right) + 2c e^{-mc^2/2RT}$$

$$\frac{d}{dc} c^2 e^{-mc^2/2RT} = -c^2 e^{-mc^2/2RT} \left(\frac{2Mc}{2RT} \right) + 2c e^{-mc^2/2RT}$$

$$\frac{d}{dc} c^2 e^{-mc^2/2RT} = 2c e^{-mc^2/2RT} - c^2 e^{-mc^2/2RT} \left(\frac{2Mc}{2RT} \right)$$

$$\frac{d}{dc} c^2 e^{-mc^2/2RT} = 2c e^{-mc^2/2RT} - c^2 e^{-mc^2/2RT} \left(\frac{M}{2RT} \right) 2c$$

$$\frac{d}{dc} c^2 e^{-mc^2/2RT} = 2c e^{-mc^2/2RT} - 2c c^2 e^{-mc^2/2RT} \left(\frac{M}{2RT} \right)$$

$$\frac{d}{dc} c^2 e^{-mc^2/2RT} = 2c e^{-mc^2/2RT} - 2c c^2 e^{-mc^2/2RT} \left(\frac{M}{2RT} \right) c^2$$

$$\frac{d}{dc} c^2 e^{-mc^2/2RT} = 2c e^{-mc^2/2RT} \left[1 - \frac{Mc^2}{2RT} \right] \quad \text{--- 3}$$

Putting the value in eq 2

$$4\pi \left[\frac{M}{2\pi R T} \right]^{3/2} \cdot 2c e^{-mc^2/2RT} \left[1 - \frac{Mc^2}{2RT} \right] = 0$$

According to mathematics:-

$$4\pi \left[\frac{M}{2\pi R} \right]^{3/2} \neq 0$$

$\Rightarrow$ Because of Molar mass can't be zero.

$$2c e^{-mc^2/2RT} \neq 0$$

$\Rightarrow$ Because c is can't be zero.

$$\Rightarrow 1 - \frac{Mc^2}{2RT} = 0$$

$$1 = \frac{Mc^2}{2RT}$$

$$2c^2 = \frac{2RT}{M}$$

$$c = \sqrt{\frac{2RT}{M}}$$

Proved.

#

Derivation of root mean square velocity C_{rms} from Maxwell's law of Distribution of velocity:-

Mean velocity of gas molecule is given as:-

$$\bar{C} = \int_0^{\infty} C f(C) dC$$

Mean square velocity of gas

molecule is given as:-

$$\bar{C}^2 = \int_0^{\infty} C^2 f(C) dC$$

Root mean square velocity of gas

molecule is given as:-

$$C_{rms} = (\bar{C}^2)^{1/2} = \left[\int_0^{\infty} C^2 f(C) dC \right]^{1/2} \quad \text{--- 1}$$

We know that:-

$$f(C) = \frac{4N}{\sqrt{2\pi R T}} \left[\frac{M}{2\pi R T} \right]^{3/2} \cdot e^{-mC^2/2RT}$$

by putting the $f(C)$ value in eq (1)

$$C_{rms} = (\bar{C}^2)^{1/2} = \left[\int_0^{\infty} C^2 \frac{4N}{\sqrt{2\pi R T}} \left[\frac{M}{2\pi R T} \right]^{3/2} \cdot e^{-mC^2/2RT} dC \right]^{1/2}$$

$$C_{rms} = (\bar{C}^2)^{1/2} = \left[\frac{4N}{\sqrt{2\pi R T}} \left(\frac{M}{2\pi R T} \right)^{3/2} \int_0^{\infty} C^4 e^{-mC^2/2RT} dC \right]^{1/2} \quad \text{--- 2}$$

$$\text{Put } \frac{M}{2RT} = a$$

$$\int_0^{\infty} C^4 e^{-mC^2/2RT} dC = \int_0^{\infty} C^4 e^{-aC^2} dC = \frac{3\sqrt{\pi}}{8a^{5/2}}$$

by putting this value in eq 2

$$C_{rms} = (\bar{C}^2)^{1/2} = \left[\frac{4N}{\sqrt{2\pi R T}} \left(\frac{M}{2\pi R T} \right)^{3/2} \cdot \frac{3\sqrt{\pi}}{8a^{5/2}} \right]^{1/2}$$

$$C_{rms} = (\bar{C}^2)^{1/2} = \left[\frac{4N}{\sqrt{2\pi R T}} \left(\frac{M}{2\pi R T} \right)^{3/2} \cdot \frac{3\sqrt{\pi}}{8} \cdot \frac{1}{a^{5/2}} \right]^{1/2} \quad \text{--- 3}$$

$$\text{Putting in eq 3 } a = \frac{M}{2RT} \Rightarrow \frac{1}{a} = \frac{2RT}{M}$$

$$C_{rms} = (\bar{C}^2)^{1/2} = \left[\frac{4N}{\sqrt{2\pi R T}} \left(\frac{M}{2\pi R T} \right)^{3/2} \cdot \frac{3\sqrt{\pi}}{8} \cdot \left(\frac{2RT}{M} \right)^{5/2} \right]^{1/2} \quad \text{--- 4}$$

$$\text{As } \left(\frac{2RT}{M} \right)^{5/2} = \left(\frac{2RT}{M} \right)^2 \cdot \left(\frac{2RT}{M} \right)^{1/2}$$

As Putting in eq 4

$$C_{rms} = (\bar{C}^2)^{1/2} = \left[\frac{4N}{\sqrt{2\pi R T}} \left(\frac{M}{2\pi R T} \right)^{3/2} \cdot \frac{3\sqrt{\pi}}{8} \cdot \left(\frac{2RT}{M} \right)^2 \cdot \left(\frac{2RT}{M} \right)^{1/2} \right]^{1/2}$$

$$C_{rms} = (\bar{C}^2)^{1/2} = \left[\frac{4N}{\sqrt{2\pi R T}} \left(\frac{M}{2\pi R T} \right)^{1/2} \cdot \frac{M}{2\pi R T} \cdot \frac{3\sqrt{\pi}}{8} \cdot \left(\frac{2RT}{M} \right)^2 \cdot \left(\frac{2RT}{M} \right)^{1/2} \right]^{1/2}$$

$$= \left[\frac{4N}{\sqrt{2\pi R T}} \cdot \frac{M}{2\pi R T} \cdot \frac{3\sqrt{\pi}}{8} \cdot \left(\frac{2RT}{M} \right)^2 \cdot \left(\frac{2RT}{M} \right)^{1/2} \right]^{1/2}$$

$$= \left[\frac{4N}{\sqrt{2\pi R T}} \cdot \frac{M}{8} \cdot \frac{3}{8} \cdot \left(\frac{2RT}{M} \right)^2 \right]^{1/2}$$

$$= \left[4\bar{h} \cdot \frac{M}{2\sqrt{\pi}} \cdot \frac{3}{8} \cdot \frac{4(RT)^{3/2}}{M^2} \right]^{1/2}$$

$$= \left[4 \cdot \frac{3}{2 \cdot 8} \cdot \frac{4RT}{M} \right]^{1/2}$$

$$= \left[\frac{12}{16} \cdot \frac{4RT}{M} \right]^{1/2} = \left[\frac{3}{4} \cdot \frac{4RT}{M} \right]^{1/2}$$

$$= \sqrt{\frac{3RT}{M}}$$

Derivation of Mean velocity $\bar{C}$ from Maxwell law of Distribution of velocity:-

Mean velocity is given as:-

$$\bar{C} = \int_0^{\infty} f(c) c \, dc \quad \text{--- 1}$$

Put $f(c) = 4\bar{h} \bar{c}^2 \left[\frac{M}{2\sqrt{\pi}} \right]^{3/2} \cdot e^{-mc^2/2RT} \cdot c \, dc$ in eq (1).

$$\bar{C} = \int_0^{\infty} 4\bar{h} \bar{c}^2 \left[\frac{M}{2\sqrt{\pi}} \right]^{3/2} \cdot e^{-mc^2/2RT} \cdot c \, dc$$

$$= \int_0^{\infty} 4\bar{h} \bar{c}^2 c \left[\frac{M}{2\sqrt{\pi}} \right]^{3/2} \cdot e^{-mc^2/2RT} \, dc$$

$$= 4\bar{h} \left[\frac{M}{2\sqrt{\pi}} \right]^{3/2} \int_0^{\infty} c^3 \cdot e^{-mc^2/2RT} \, dc \quad \text{--- 2}$$

$$\int_0^{\infty} c^3 e^{-mc^2/2RT} \, dc = 2 \left(\frac{RT}{M} \right)^2$$

Put the value in eq (2),

$$\bar{C} = 4\bar{h} \left(\frac{M}{2\sqrt{\pi}} \right)^{3/2} \cdot 2 \left(\frac{RT}{M} \right)^2$$

$$\bar{C} = 8\bar{h} \left[\frac{M}{2\sqrt{\pi}} \right]^{3/2} \cdot \left(\frac{RT}{M} \right)^2 \quad \text{--- 3}$$

Put $\left[\frac{M}{2\sqrt{\pi}} \right]^{3/2} = \left(\frac{M}{2\sqrt{\pi}} \right)^{1/2} \cdot \frac{M}{2\sqrt{\pi}}$ in eq 3

$$\bar{C} = 8\bar{h} \left(\frac{M}{2\sqrt{\pi}} \right)^{1/2} \cdot \left(\frac{M}{2\sqrt{\pi}} \right) \cdot \frac{RT \times RT}{M \times M}$$

$$= 8\bar{h} \left[\frac{M}{2\sqrt{\pi}} \right]^{1/2} \cdot \frac{RT}{2\sqrt{\pi}} \quad \text{--- 4}$$

Put $\frac{RT}{2\sqrt{\pi}} = \left[\frac{RT}{2\sqrt{\pi}} \right]^{1/2} \cdot \left[\frac{RT}{2\sqrt{\pi}} \right]^{1/2}$ in eq 4

$$= 8\bar{h} \left[\frac{M}{2\sqrt{\pi}} \right]^{1/2} \cdot \left[\frac{RT}{2\sqrt{\pi}} \right]^{1/2} \cdot \left[\frac{RT}{2\sqrt{\pi}} \right]^{1/2}$$

$$= 8\bar{h} \cdot \left(\frac{RT}{2\sqrt{\pi}} \right)^{1/2} \cdot \left(\frac{RT}{2\sqrt{\pi}} \right)^{1/2} \cdot \left(\frac{RT}{2\sqrt{\pi}} \right)^{1/2} \quad \text{--- 5}$$

Put $(2\sqrt{\pi})^{1/2} \cdot (2\sqrt{\pi})^{1/2} = 2\sqrt{\pi}$ in eq 5

$$= 8\bar{h} \cdot \frac{(RT)^{1/2}}{2\sqrt{\pi}} \cdot (2\sqrt{\pi})^{1/2}$$

$$= 8\bar{h} \cdot \frac{(RT)^{1/2}}{2\sqrt{\pi}} \cdot (2\sqrt{\pi})^{1/2} \cdot (M)^{1/2} \quad \text{--- 6}$$

Put $\left[\frac{RT}{M} \right]$ in eq 6

$$= 8\bar{h} \cdot \frac{RT}{2\sqrt{\pi}} \cdot (2\sqrt{\pi})^{1/2} \cdot \frac{RT}{M}$$

$$\begin{aligned}
 &= \frac{4}{(2\pi)^{1/2}} \cdot \sqrt{\frac{R_1}{M}} \\
 &= \frac{4}{\sqrt{2\pi}} \cdot \sqrt{\frac{R_1}{M}} \\
 &= \frac{4}{\sqrt{2}} \cdot \sqrt{\frac{R_1}{M}} \\
 \text{Put } \frac{4}{\sqrt{2}} &= \sqrt{8} \\
 &= \sqrt{8} \cdot \sqrt{\frac{R_1}{M}} \\
 &= \sqrt{8R_1} \\
 &= \sqrt{\frac{8R_1}{M}}
 \end{aligned}$$

#

Max Boltzmann law of distribution of velocities:-

This law was proposed by James Clerk Maxwell, a Scottish Physicist and Ludwig Edward Boltzmann (An Austrian Physicist).

Statement:-

- 1) A Particular fraction of gas molecules having Particular velocity range is Constant.
- 2) According to Maxwell's Law, distribution of velocities of gas molecules depends on temperature & molecular mass.

Assumption:-

- 1) No. of gas molecules in a system remains constant during observed process.
- 2) Total energy of gas system remains constant during observed process.
- 3) Temperature of gas system remains constant during observed process.
- 4) Gas molecules are distinguished.
- 5) There is no restriction on movement of gas molecules. i.e., gas molecules may be accommodated in

any energy level.

∴ No. of gas molecules having velocities in the range of " c " and " $c+dc$ " is constant.

Suppose:-

$$C = 100 \text{ m s}^{-1}$$

$$dc = 0.1 \text{ m s}^{-1}$$

$$c + dc = 100 + 0.1 = 100.1 \text{ m s}^{-1}$$

→ This phenomenon is steady state.

In this phenomenon, no. of gas molecules facing increase in velocity is equal to no. of molecules facing decrease in velocity by some amount.

⇒ Fraction of gas molecules having velocities in the range of c and $c+dc$ is constant.

⇒ No. of gas molecules dN having specific velocity in the range of c and $c+dc$ is constant at

constant temperature.

⇒ Probability of gas molecules $\frac{1}{N} \cdot dN$ or $f(c)$ having specific velocity in the range of c and $c+dc$ is constant at constant temperature.

Fraction of gas molecules having velocities in the range of c and $c+dc$ is constant.

$$\frac{dN}{N} = f(c) \cdot dc$$

$$\frac{dN}{N} = f(c) \cdot dc = \frac{e^{-E/K}}{\int_0^{\infty} e^{-E/K} dx}$$

K = Boltzmann constant

C_x - velocity of gas molecules along x -axis.

Energy of gas molecules can be

written as:-

$$E = \frac{1}{2} m v^2$$

velocity of gas molecules along x -axis:

$$E = \frac{1}{2} m C_x^2 = \frac{m C_x^2}{2}$$

Put this value in eq 1

$$\frac{dN}{N} = f(c) \cdot dc = \frac{e^{-m C_x^2 / 2K}}{\int_0^{\infty} e^{-m C_x^2 / 2K} \cdot dx}$$

As, no. of molecule having velocity b/w

$-\infty$ to 0 are equal to no. of

molecules having velocity b/w 0 to $+\infty$.

$$\int_{-\infty}^0 e^{-m C_x^2 / 2K} \cdot dx = \int_0^{\infty} e^{-m C_x^2 / 2K} \cdot dx + \int_0^{\infty} e^{-m C_x^2 / 2K} \cdot dx$$

$$\int_{-\infty}^{\infty} e^{-mc^2/2kT} \cdot dx + \int_0^{\infty} e^{-mc^2/2kT} = \int_0^{\infty} e^{-mc^2/2kT} dx + \int_0^{\infty} e^{-mc^2/2kT} dx$$

$$\int_{-\infty}^{\infty} e^{-mc^2/2kT} \cdot dx + \int_0^{\infty} e^{-mc^2/2kT} = 2 \int_0^{\infty} e^{-mc^2/2kT} \cdot dx$$

Putting these value in eq 2

$$\frac{dN}{N} = f(c) dx = \frac{e^{-mc^2/2kT}}{2 \int_0^{\infty} e^{-mc^2/2kT} \cdot dx} \cdot dx \quad \text{--- 3}$$

For Simplify:

$$a = \frac{m}{2kT}$$

eq. 3 becomes:

$$\frac{dN}{N} = f(c) dx = \frac{e^{-ac^2}}{2 \int_0^{\infty} e^{-ac^2} \cdot dx} \cdot dx \quad \text{--- 4}$$

$$\int_0^{\infty} e^{-ac^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{a}}$$

Put the value in eq 4-

$$\frac{dN}{N} = f(c) dx = \frac{e^{-ac^2}}{2 \times \frac{1}{2} \sqrt{\frac{\pi}{a}}} \cdot dx$$

$$\frac{dN}{N} = f(c) dx = \frac{e^{-ac^2} \cdot dx}{\sqrt{\frac{\pi}{a}}}$$

$$= \frac{e^{-ac^2} \cdot dx}{\sqrt{\frac{\pi}{a}}}$$

$$= e^{-ac^2} dx \cdot \sqrt{\frac{a}{\pi}}$$

By putting the value of $a = \frac{m}{2kT}$

$$= \sqrt{\frac{m}{2kT}} \cdot e^{-\frac{mc^2}{2kT}} \cdot dx \quad \text{--- 5}$$

As we know:

$$\text{Molar mass (M)} = \text{molecular mass (m)} \times N_A$$

$$M = m \times N_A$$

$$\therefore m = \frac{M}{N_A}$$

by putting the value in eq 5

$$\frac{dN}{N} = f(c) dx = \sqrt{\frac{M}{2kT N_A}} \cdot e^{-\frac{mc^2}{2kT}} \cdot dx$$

$$\therefore \text{Put } R = k N_A$$

$$\frac{dN}{N} = f(c) dx = \sqrt{\frac{M}{2RT}} \cdot e^{-\frac{mc^2}{2kT}} \cdot dx \quad \text{--- 6}$$

For three dimension eq 6 will become:

$$\frac{dN}{N} = f(c_x, c_y, c_z) \cdot dx dy dz = \left(\frac{M}{2RT} \right)^{3/2} \cdot e^{-\frac{m(c_x^2 + c_y^2 + c_z^2)}{2kT}} \cdot dx dy dz \quad \text{--- 7}$$

$$\text{Put: } C_x C_y C_z = c$$

$$dx dy dz = dc$$

eq 7 will become:

$$\frac{dN}{N} = f(c) \cdot dc = \left(\frac{M}{2RT} \right)^{3/2} \cdot e^{-\frac{mc^2}{2kT}} \cdot dc \quad \text{--- 8}$$

eq 8 shows fraction of gas molecules in spherical shell thickness dc .

∴ Volume of sphere = $\frac{4}{3}\pi r^3$

Volume of Spherical Shell = $dv = 4\pi r^2$

$$[C=C]$$

$$dv = 4\pi r^2$$

by multiply with $4\pi r^2$ of eq 8

$$\frac{dN}{N} = f(c) \cdot dc = 4\pi n^2 \left[\frac{M}{2\pi R T} \right]^{3/2} \cdot e^{-\frac{mc^2}{2RT}} \cdot dc$$

∴ by "de" on B.S:-

$$\frac{dN}{N} \cdot \frac{1}{dc} = f(c) \cdot dc = 4\pi n^2 \left[\frac{M}{2\pi R T} \right]^{3/2} \cdot e^{-\frac{mc^2}{2RT}} \cdot dc$$

$$\frac{dN}{N} \cdot \frac{1}{dc} = f(c) = 4\pi n^2 \left[\frac{M}{2\pi R T} \right]^{3/2} \cdot e^{-\frac{mc^2}{2RT}}$$

Limitations:-

- Maxwell was not convince of his own work.
- He Paid no attention to mechanism of Collision.
- Maxwell's law assumes C_x, C_y and C_z are independent of each other.
- Maxwell's law does not hold at extremely low temperature and at very high density.

#

Maxwell's Law of Distribution of energies:-

→ Maxwell's Law of Distribution

of velocities is given as:-

$$\frac{1}{N} \cdot \frac{dN}{dc} = f(c) = 4\pi n^2 \left[\frac{M}{2\pi R T} \right]^{3/2} \cdot e^{-\frac{mc^2}{2RT}} \quad \text{--- 9}$$

$\frac{1}{N} \cdot \frac{dN}{dc}$ = Probability of gas molecules.

$$\text{Putting } E = \frac{1}{2} m c^2$$

$$M c^2 = 2 E \quad \text{--- 2}$$

By Differentiating with respect to "c".

$$M \cdot \frac{d}{dc} c^2 = 2 \frac{dE}{dc}$$

$$2CM = 2 \frac{dE}{dc}$$

$$MC = \frac{dE}{dc}$$

$$MC \cdot dc = dE \quad \text{--- 3}$$

Taking square root of eq 3 in B.S

$$\sqrt{MC^2} = \sqrt{2E}$$

$$CM = \sqrt{2E} \quad \text{--- 4}$$

$$C^2 \propto E \quad \text{eq 3 & 4}$$

$$MC \cdot dc \times CM = dE \times \sqrt{2E}$$

$$M \cdot \sqrt{M} \cdot c^2 dc = dE \sqrt{E}$$

$$\therefore M \sqrt{M} = (m)^{3/2}$$

$$m^{3/2} \cdot c^2 dc = \sqrt{2E} dE$$

$$c^2 \cdot dc = \frac{\sqrt{2E} dE}{(m)^{3/2}} \quad \text{--- } \int$$

According to:

$$\frac{1}{N} \cdot \frac{dN}{dc} f(c) = 4\pi \bar{n} \left[\frac{M}{2\pi R T} \right]^{3/2} \cdot e^{-m^2/2\pi R T}$$

by Rearrange eq 1

$$\frac{dN}{N} = f(c) = 4\pi \bar{n} \left(\frac{M}{2\pi R T} \right)^{3/2} \cdot e^{-m^2/2\pi R T} \cdot c^2 dc$$

→ Put value of m^2 from eq 2 and

$c^2 dc$ from eq 5.

$$\frac{dN}{N} = f(c) = 4\pi \bar{n} \left(\frac{M}{2\pi R T} \right)^{3/2} \cdot e^{-E/2\pi R T} \cdot \sqrt{2E} \cdot dE$$

$$\frac{dN}{N} = f(c) = 4\pi \bar{n} \frac{(M)^{3/2}}{(2\pi R T)^{3/2}} \cdot e^{-E/2\pi R T} \cdot \sqrt{2E} \cdot dE$$

$$= 4\pi \bar{n} \cdot e^{-E/2\pi R T} \cdot \sqrt{2E} \cdot dE$$

$$(2\pi R T)^{3/2}$$

$$= \frac{4\pi \bar{n}}{(2\pi R T)^{3/2}} \cdot e^{-E/2\pi R T} \cdot \sqrt{2E} \cdot dE$$

$$= \frac{4\pi \bar{n}}{(2\pi R T)^{3/2}} \cdot e^{-E/2\pi R T} \cdot \sqrt{2E} \cdot dE$$

$$4\pi \bar{n} \cdot (2\pi R T)^{3/2}$$

$$\frac{dN}{N} = f(c) = \frac{4\pi \bar{n}}{(2\pi R T)^{3/2}} \cdot e^{-E/2\pi R T} \cdot \sqrt{2E} \cdot dE$$

$$dN = f(c) = \frac{4\pi \bar{n}}{(2\pi R T)^{3/2}} \cdot e^{-E/2\pi R T} \cdot \sqrt{2E} \cdot dE$$

→ Distribution of energy is same as the distribution of velocity.